

Nanosculpting antiferromagnets to explore novel electronic transport

PhD opportunity · University of Warwick · Department of Physics · Starting October 2027

Supervisor: Dr Max Birch · Royal Society University Research Fellow
Superconductivity and Magnetism Group · Nanosculpted Quantum Materials

Electronic functionality from unconventional antiferromagnets

Can we sculpt antiferromagnetic materials into devices to reveal new physics and control their emergent behaviour? Antiferromagnets contain ordered magnetic moments that typically cancel one another. However, certain unconventional antiferromagnets exhibit electronic responses associated with broken time-reversal symmetry (TRS) (usually associated with ferromagnets), opening possibilities for new spintronic devices.

This experimental PhD project explores two families of these materials. In noncollinear antiferromagnets [1,2], intricate arrangements of magnetic moments can form magnetic multipole structures, such as octupoles, that activate unexpected behaviour despite a near-zero net magnetisation (Fig. 1a). In altermagnets [3,4], the interplay between antiparallel magnetic spins and their crystal environments produces an electronic spin splitting that changes with direction in momentum space (Fig. 1b). These materials offer opportunities to connect fundamental questions about magnetic symmetry with electronic device functionality.

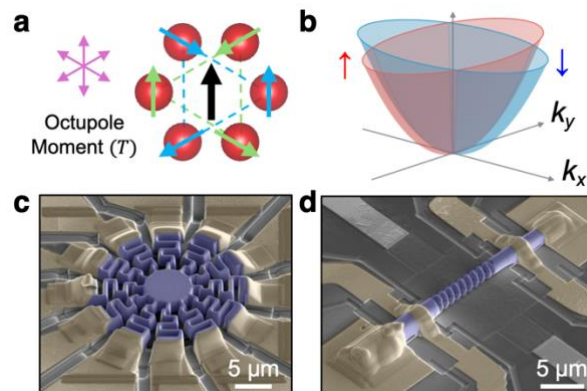


Fig. 1

Sculpting materials into intricate devices

You will use focused ion beam (FIB) fabrication to create intricate microscopic devices directly from furnace-grown crystalline materials (examples in Fig. 1c,d). By designing their geometry, electrodes and mechanical structures, we can probe and influence material behaviour in ways that conventional samples cannot readily accommodate. The project will use this freedom to ask creative experimental questions. Can we realise and electrically probe metastable TRS-breaking antiferromagnetic states? Can we create antiferromagnetic diode responses, where the device resistance depends on current direction? Can mechanical deformation control the antiferromagnetic order and its transport signatures? The specific experiments will develop with the results and your interests, with opportunities to design your own devices and pursue unexpected discoveries.

Training and research environment

You will receive direct training from Dr Max Birch, who specialises in FIB fabrication of quantum-material devices [5-9], and develop expertise in electrical transport measurements, including nonlinear measurements using lock-in amplifiers. You will learn how to turn a new idea into an experiment, uncover the physics behind the results, and use that understanding to control the material's behaviour. Within Warwick's Superconductivity and Magnetism Group, you will have access to cryomagnet systems and opportunities to collaborate with crystal growers. The core fabrication and measurement facilities are expected to be ready before you arrive, with opportunities to develop new experimental capabilities as the project progresses. International connections offer potential collaborations in Japan, the USA and Germany, and opportunities to use synchrotron x-ray and neutron facilities where relevant to the project.

Who we are looking for

We are looking for someone who enjoys experimental physics, working with their hands and finding creative solutions to practical problems. Fabricating these devices involves carefully sculpting and assembling microscopic structures inside the focused ion beam instrument with a micromanipulator, rewarding patience and attention to detail. In return, the flexibility of the approach provides extreme freedom to design new devices and experiments, turning imaginative scientific questions into structures you can investigate. You do not need previous focused ion beam experience: specialist training and guidance will be provided.

If you are interested, please contact Dr Max Birch to discuss the project.

Email: maximilian.birch@riken.jp · **Group information:** nanosculpt.github.io · **Apply:** [here](#)

[1] S. Nakatsuji, *et al.* Nature **527**, 212–215 (2015). [DOI](#). [2] B. H. Rimmler, B. Pal and S. S. P. Parkin, Nat. Rev. Mater. **10**, 109–127 (2025). [DOI](#). [3] L. Šmejkal, J. Sinova and T. Jungwirth, Phys. Rev. X **12**, 040501 (2022). [DOI](#). [4] C. Song *et al.*, Nat. Rev. Mater. **10**, 473–485 (2025). [DOI](#). [5] M. T. Birch *et al.*, Nat. Commun. **11**, 1726 (2020). [DOI](#). [6] M. T. Birch *et al.*, Nat. Commun. **13**, 3630 (2022). [DOI](#). [7] M. T. Birch *et al.*, Nature **633**, 554–559 (2024). [DOI](#). [8] M. T. Birch *et al.*, Nat. Nanotechnol. **21**, 352–358 (2026). [DOI](#). [9] M. T. Birch *et al.*, PRB **114**, 094409 (2026). [DOI](#).